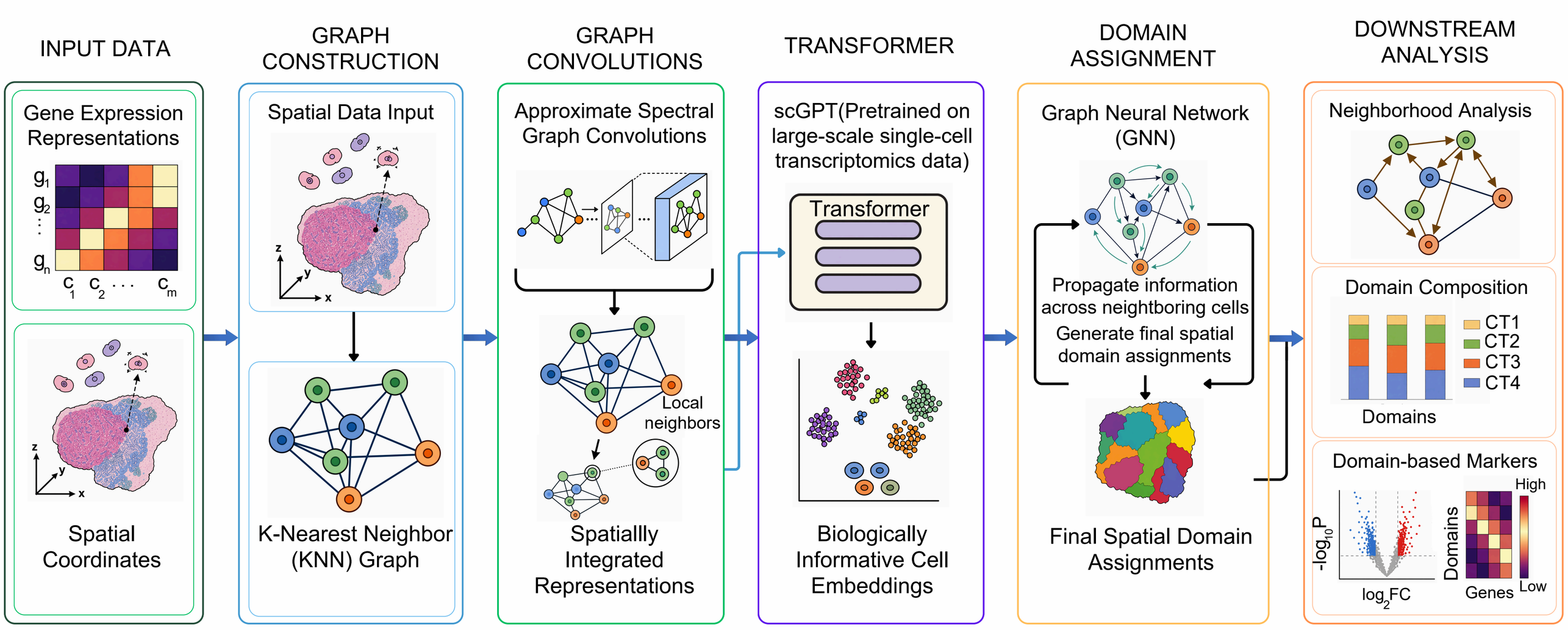
INPUT DATA
GRAPH CONSTRUCTION
GRAPH CONVOLUTIONS
TRANSFORMER
DOMAIN ASSIGNMENT
DOWNSTREAM ANALYSIS
Gene Expression Representations
Spatial Coordinates
Spatial Data Input
K-Nearest Neighbor (KNN) Graph
Approximate Spectral Graph Convolutions
Local neighbors
Spatiallly Integrated Representations
scGPT(Pretrained on large-scale single-cell transcriptomics data)
Transformer
Biologically Informative Cell Embeddings
Graph Neural Network (GNN)
Propagate information across neighboring cells
Generate final spatial domain assignments
Final Spatial Domain Assignments
Neighborhood Analysis
Domain Composition
CT1
CT2
CT3
CT4
Domains
Domain-based Markers
-log10P
log2FC
Domains
Genes
High
Low